



AI-Enhanced Digital Picture Frame to Improve Older Adult Social & Mental Health



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Project Definition

Problem

Technology is advancing at a faster rate than ever before, and it's not slowing down any time soon. With 7 billion people owning mobile devices, the modern reliance on digital technology is something the older adult population had never experienced growing up. As a result, their inexperience with such technology is separating them from the rest of the digital world behind what's known as the "digital divide", which is detrimental to older adults' mental and social health. According to BioMed Central Health, 50% of people over the age of 60 are at risk of social isolation (Fakoya). A survey conducted by the University of Michigan in 2023 found that $\frac{1}{3}$ of US adults ages 50-80 feel socially isolated (Malani). Additionally, loneliness and social isolation increase the risk of other dangerous risk factors, including heart disease, diabetes, stroke, and earlier death ("Health Effects"). Kept up, these factors will continue to degrade seniors' social, mental, and physical health as time goes on.

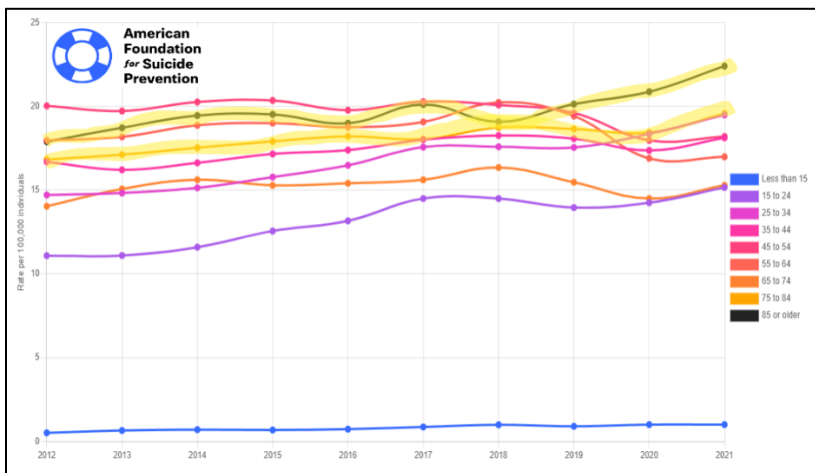


Figure 1: suicide rates by age group, 2012-2021; ages 75+ are highlighted in yellow with the highest rates

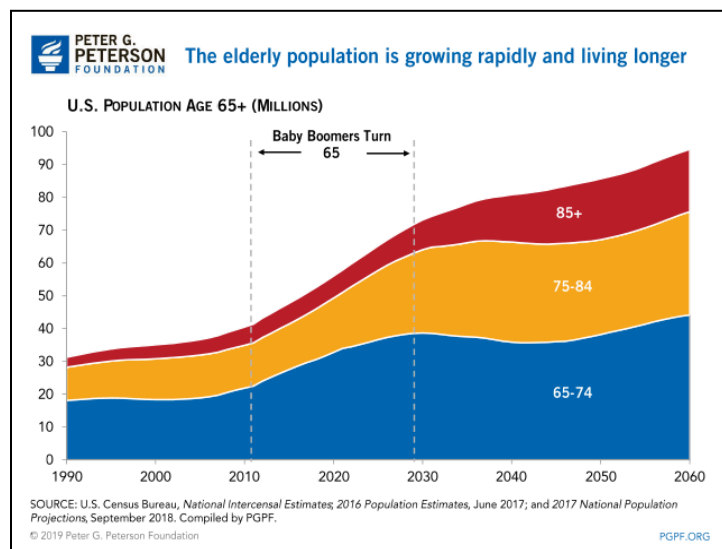


Figure 2: projected US elderly population, 1990-2060

Furthermore, older adults ages 75 and older have the highest proportion of suicides of any age group, according to the American Foundation for Suicide Prevention (Fig. 1). Not only that, but the older adult population is living longer and consequently growing larger

due to advances in technology. They are projected by the Peterson Foundation to nearly double in size by 2060 from 55 million to 95 million (Fig. 2). This means that in the future, more older adults will be subject to this problem for longer if this problem is not stopped now.

Solution/Project Proposal

We had conversations with many older adults, their loved ones, and their caretakers to identify their values in a solution to this problem. Their four main values were socialization, mental stimulation, engagement, and easy usage. With these values in mind, we designed an AI-Enhanced Digital Picture Frame to improve older adults' mental and social health, the FamilyFrame. It uses artificial intelligence to generate tailored conversation prompts for users based on their inputted information, which adds a fun, personalized, and unique twist to their conversations to enhance socialization. The FamilyFrame also provides personalized fun facts to keep users engaged and mentally active. Additionally, the FamilyFrame serves as a digital picture frame when not in use to shuffle through heartwarming pictures of whatever the user uploads to it. Finally, with its minimalistic interface featuring large buttons and text, the FamilyFrame is designed to be incredibly intuitive and easy to use regardless of a user's technological experience. All of these features combined will surely improve older adults' relationships with their loved ones and keep them mentally and socially healthy. Users can either download the FamilyFrame app for purchase online or buy the physical FamilyFrame tablet with the software and case. The FamilyFrame is produced by pairing our FamilyFrame software with ChatGPT tokens in a refurbished tablet, all of which yield relatively low production costs. This results in the FamilyFrame being a profitable product for our company and an effective solution for the beloved older adults in our community.

Target Dates

- 07/25/2023 - Solution Discovered
- 08/11/2023 - Concept Generations Due
- 09/07/2023 - 5 Concepts Selected
- 10/04/2023 - Final Concept Selected
- 10/27/2023 - Samsung Solve For Tomorrow Competition
- 12/07/2023 - Thomas Edison Pitch Competition Video 1 Due
- 01/05/2024 - Mid-Year Engineering Posters Due
- 01/25/2024 - Mid-Year Presentations
- 04/10/2024 - Thomas Edison Pitch Competition Video 2 Due
- 05/03/2024 - Income | Cost Spreadsheet
- 05/03/2024 - Business Canvas
- 05/03/2024 - Customer Discovery Sheet
- 05/22/2024 - Siena Final Presentations
- 06/13/2024 - Completion of Prototype
- 06/13/2024 - Final Engineering Poster Due
- 06/21/2024 - Final Presentations

Project Team

Edward Xu - Chief Executive Officer (CEO)

Jerome Rodrigo - Chief Financial Officer (CFO)

Kevin Smith - Capstone Advisor, Technology Teacher at Shaker High School

Christopher VanDerbeck - Business Mentor, Business Teacher at Shaker High School

Joan Rueckert - Project Mentor, Outreach Worker at Colonie Village Senior Center

Anthony DeBonis - Project Mentor, President of Troy Web Consulting

Shane Hoffman - Project Mentor, Developer at Troy Web Consulting

Joseph Iaquinto - Project Mentor, Senior Developer at Troy Web Consulting

Tyler Flood - Project Mentor, Senior Visual Designer at Troy Web Consulting

Concept Generation

Interestingly enough, our project did not start with this idea and instead with something totally different: automatic sheet music turning. We struggled to find a problem we were both passionate to solve and therefore started off the year with this problem as we were both musicians that often struggled with turning pages while playing music. Of course, we ended up pivoting from this idea completely after discussing it with our Capstone Advisor, Mr. Smith. Nevertheless, our twenty concepts for automatic sheet music turning are listed below.

First Twenty Concepts

1. Single Mechanical Arm with Wheel
2. Multiple Mechanical Arms Top Attachment
3. Multiple Mechanical Arms Bottom Attachment
4. Multiple Slotted Mechanical Arms Top Attachment
5. Multiple Slotted Mechanical Arms Bottom Attachment
6. Clip with Spacers
7. Mechanical Arm and Spacer Clip
8. Mechanical Arm and Magnetic Spacers
9. Mechanical Arm and Air Suction Piece
10. Adhesive Mechanical Arm
11. Multiple Slotted Mechanical Arms Wired Bottom Attachment
12. Suction Cup Top Attachment
13. Suction Cup Bottom Attachment
14. Compressed Air Nozzle
15. Multi-Arm with Clips Top Attachment
16. Multi-Arm with Clips Bottom Attachment
17. Static Electricity Turning Rod Top Attachment
18. Static Electricity Turning Rod Bottom Attachment
19. Side Arm Page Turner
20. Mechanical Arm and Static Rod

Concept Selection

When we started generating concepts for our real problem, we knew we wanted to utilize the power of artificial intelligence to give the user a truly personalized experience, so all our concepts share that similarity. Given our shortened brainstorming process due to our idea pivot, we are presenting fewer competitive concepts (three instead of five).

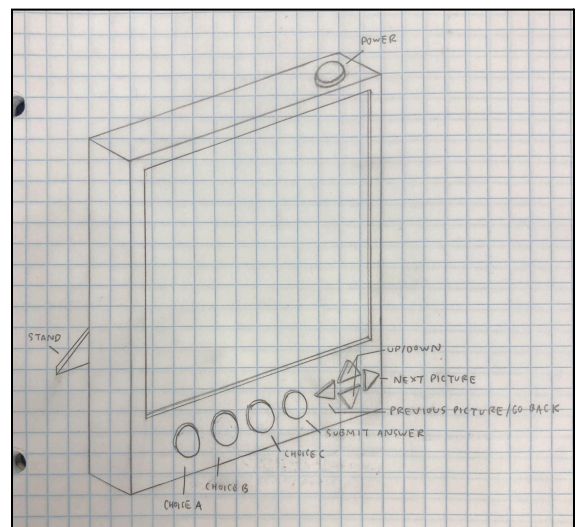
Description of Competitive Concepts



Teddy Bear AI
Conversation Assistant

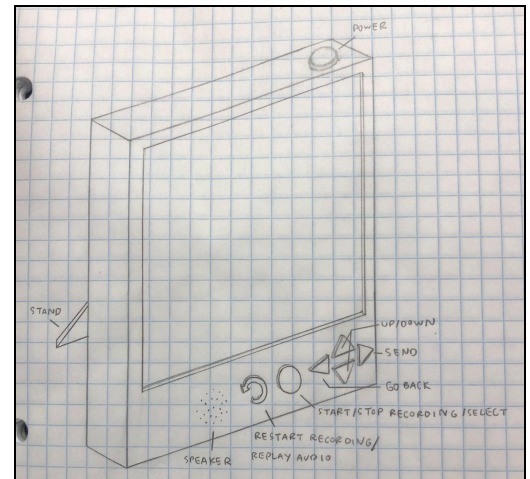
Our first competitive concept was a Teddy Bear AI Conversation Assistant. Its main job was to generate conversation prompts for older adults to discuss in voice messages, which would be sent through the teddy bear to a loved one's smart device. The loved one could then send voice messages back through the teddy bear. With built-in buttons for this job, the teddy bear is a cute and cuddly product for older adults to have engaging conversations with their loved ones. However, it doesn't offer anything beyond this feature that we know are important as well, such as mental stimulation.

On the other hand, our second competitive concept, the Digital Picture Frame w/ AI-Generated MCQs, focuses on mental stimulation. It processes user-inputted information to generate tailored multiple-choice questions with pictures for the user to answer about their experiences, interests, and memories. Contrary to the teddy bear, although it provides mental stimulation, it does not provide social stimulation to the user.



Digital Picture Frame w/
AI-Generated MCQs

Finally, our third competitive concept was a Picture Frame AI Conversation Assistant. It essentially combines the first and second concepts into one, offering the AI Conversation Assistant feature of the teddy bear with the picture frame structure of the digital picture frame. This was the basis of the concept we went with to design our FamilyFrame, which we modified by adding tailored fun facts and removing the physical buttons.



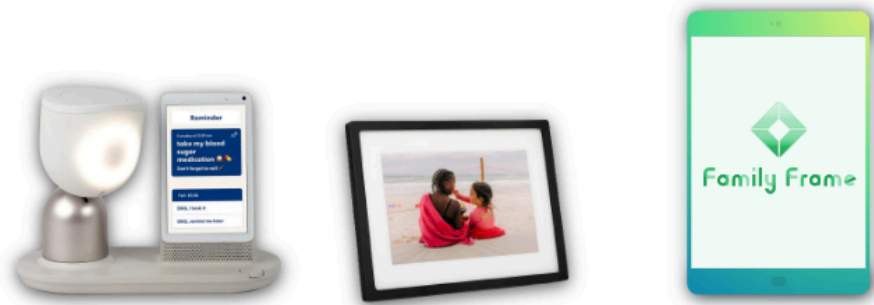
Picture Frame AI
Conversation Assistant

Evaluation Matrix

Evaluation Matrix				
Criteria	Weight	Teddy Bear AI Conversation Assistant	Picture Frame AI Conversation Assistant	Digital Picture Frame w/ AI MCQs
Ease of Setup/Usage	4	4	3	2
Ease of Manufacturing	2	3.5	3	2.5
Cost of Materials	1	4.5	3	3
Mental Stimulation	5	2	2.5	5
Social Stimulation	5	5	5	1
Durability	3	3	3	3
Engagement	4	3	5	4
Overall Rating	-	83.5	87.5	71
*Weight is on a 1-5 scale, with 5 being given the highest importance.				

To properly compare our three competitive concepts, we met with one of our project mentors Anthony DeBonis, President of Troy Web Consulting, to construct this Evaluation Matrix with the above seven criteria and weightings. From there, we scored each concept in each category and determined that the Picture Frame AI Conversation Assistant was the best concept. It had the best balance between both mental and social stimulation as it combined features from both of the other two concepts, resulting in it having the highest engagement score as well. These factors are very important to us as we discovered in our customer discovery, so we determined this design to be the one we would proceed with developing.

Product Competitors



	ELLIQ	SKYLIGHT FRAME	FAMILYFRAME
SOCIALIZATION	—	✗	✓
PICTURE FRAME	✗	✓	✓
MENTAL STIMULATION	✓	✗	✓
COST	\$250 + \$60/MONTH	\$160 OR \$300	\$150 + \$3/MONTH

When it comes to direct competitors, there's virtually no existing product out there comparable to the FamilyFrame. The two closest competitors we could find through extensive research were the ElliQ (left) and Skylight Frame (middle).

The ElliQ is a voice-activated smart assistant with a moving lamp that acts as a companion for older adults. While it does provide mental stimulation to its users through engaging activities, it more so replaces human interaction with itself rather than enhances human interaction like the FamilyFrame does. Additionally, besides the moving lamp companion aspect, the ElliQ's abilities do not extend much further than those of your typical smart device. And for a whopping \$250 enrollment fee and \$60 monthly subscription, it'll have you paying nearly \$1000 just for your first year of using it.

The Skylight Frame on the other hand is purely a stand-alone digital picture frame that works in the same way our FamilyFrame does in its picture frame mode. However, its price at either \$160 or \$300 with an optional \$39 yearly subscription for premium features puts it at nearly the same price level of the FamilyFrame—with only a fraction of the capabilities.

Project Proposal

Our proposed solution is the FamilyFrame, an AI-Enhanced Digital Picture Frame designed to help older adults stay connected with their loved ones and reduce their risk of social isolation and loneliness. It is a user-friendly communication device and digital picture frame that utilizes artificial intelligence to generate personalized conversation prompts and display heartwarming pictures and fun facts. The FamilyFrame is designed with simplicity and practicality to reduce as many technological difficulties as possible, and its colors were contrast checked to ensure trouble-free usage for those with impaired vision.

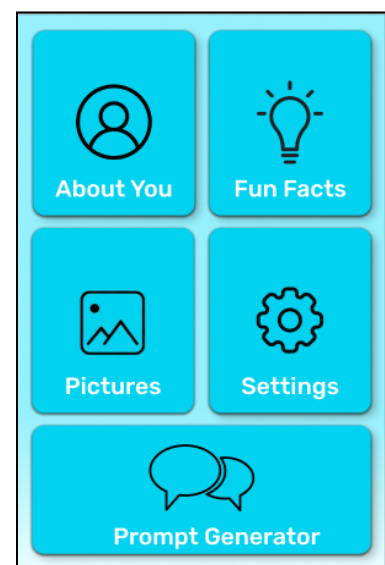
Final Design Prototype Subsystems

AI - ChatGPT-3.5 Turbo API

We use AI as a tool to enhance human interaction rather than replace it. The FamilyFrame mainly uses AI to generate conversation prompts from user-inputted data about the older adult and their loved ones for them to easily relate to and talk about. AI is also used to provide fun facts tailored to the older adults' interests. The FamilyFrame and the app will incorporate the ChatGPT-3.5 Turbo API to perform these tasks.

Home Screen

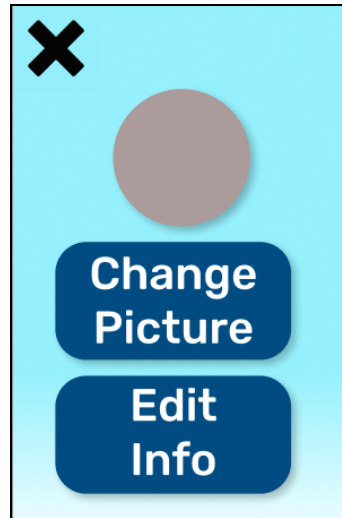
These simple five buttons give the user everything they need to easily navigate the FamilyFrame. From the Home Screen, users can fill out information about themselves, read a fun fact, enter picture frame mode, or generate conversation prompts.



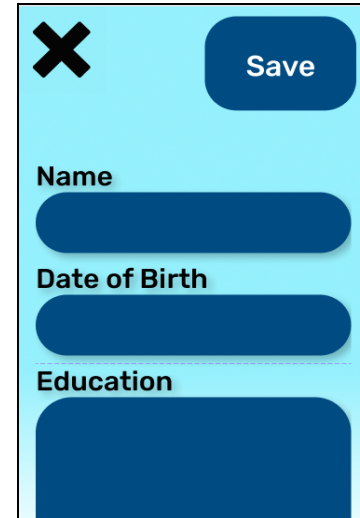
Home Screen

About You

Here, the user can fill out and edit their information for the AI to analyze when generating personalized conversation prompts and fun facts for them. They can add a profile picture and input their date of birth, education, memories, experiences, hobbies, and interests.



About You



Edit Info

Fun Fact

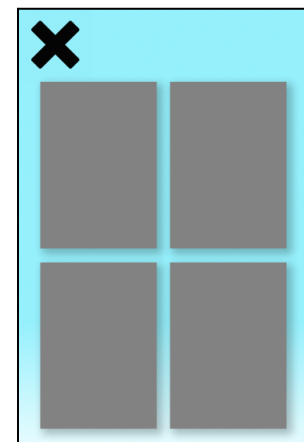
Personalized fun facts can be accessed and regenerated in this tab whenever desired. Here is an example of a fun fact generated by the AI that a chess-loving user might see.



Fun Facts

Pictures

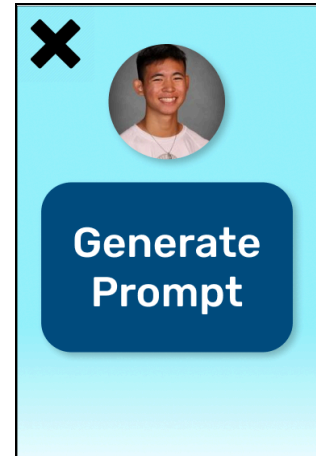
Here is where users upload and share their favorite memories with each other. These pictures can be manually displayed at any time and will also automatically start shuffling after a designated idle time to enter "picture frame mode".



Pictures

Prompt Generator

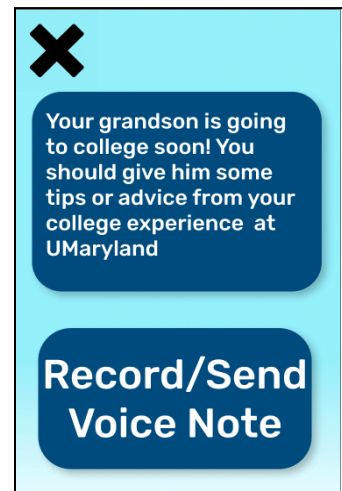
Here is where users can generate conversation prompts to start a conversation with their loved ones. These prompts are based on information that the users can easily relate to discuss. These prompts will not only exercise older adults' mental health by asking them to recall past memories and experiences, but their social health too by enhancing their conversations when discussing these prompts.



Prompt Generator

Example Prompt

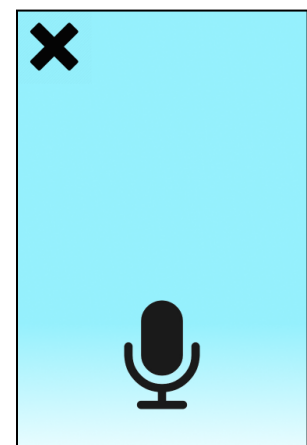
Here's an example of a prompt that the AI might generate for Edward's grandfather to discuss with Edward. The AI recognizes that Edward is going to college and asks his grandfather to give Edward some advice, knowing that his grandfather is a college alumni.



Example Prompt

Record/Send Voice Note

Recording and sending voice notes has never been easier with the use of a few large buttons.



Record Voice Note

Preliminary Bill of Materials

Prototype Costs	
Refurbished Tablet (Fire Tablet)	\$40.00
3D Printed Photo Frame Case	\$10.00
AI Costs - ChatGPT Tokens	\$0.25/month
Neodymium Disc Magnets	\$8.00
Total	\$58.00 + \$0.25/month

Fabrication & Assembly

The FamilyFrame consists of a tablet encased in a custom 3D printed case held together by magnets. The magnets—8 neodymium disk magnets (4 on each side of the case)—allow the two parts of the case to easily snap into place and be removed just as easily. When buying the physical FamilyFrame, the software will be pre-installed onto the tablet.

Customer Needs

Although the FamilyFrame is meant to improve mental and social health among older adults, our customer base extends beyond that population. In fact, older adults are more so our end users rather than our customers. Our primary customer base relies mostly on older adults' loved ones and caretakers to buy the FamilyFrame for their older adults. This includes older adults' families and friends as well as senior center workers for those that live in senior living communities.

Customer Survey

To determine our customer needs, we made phone calls and met virtually with many workers in the field of elderly care and older adults' loved ones to hear about their experiences and perspectives. Some notable individuals we talked to were Joan Rueckert, an Outreach Worker at Colonie Village Senior Center, and Anthony DeBonis, the president of Troy Web Consulting. Joan Rueckert noted that the residents at her senior center wanted a more engaging activity for them to participate in. She said that the current activities they have, like bingo, are not satisfactory enough for her residents, and that they want something more mentally and/or socially stimulating. Mr. DeBonis, as an experienced software developer and caretaker of his mother, emphasized the importance of a user-friendly interface to allow for an easier and more engaging experience for older adult users.

After considering this insight with the rest of our conversations, we determined that the values of our customers boiled down to essentially four points: socialization, mental stimulation, engagement, and easy usage. As a result, we established these values as the four pillars of our solution and centered the FamilyFrame's design around them.

Study of Current Market Conditions

Identification of Competitive Quantitative Performance

Our target market includes older adults in the US and their caretakers. Although messaging platforms already exist for them to use, they are simply harder for older adults to use due to the digital divide, and aren't nearly as engaging as our FamilyFrame. Other products don't enhance socialization like we do, but rather replace it. The FamilyFrame helps older adults easily bridge the digital divide to connect to their loved ones with a more enhanced and engaging experience thanks to our implementation of artificial intelligence.

Business Plan/Economic Analysis

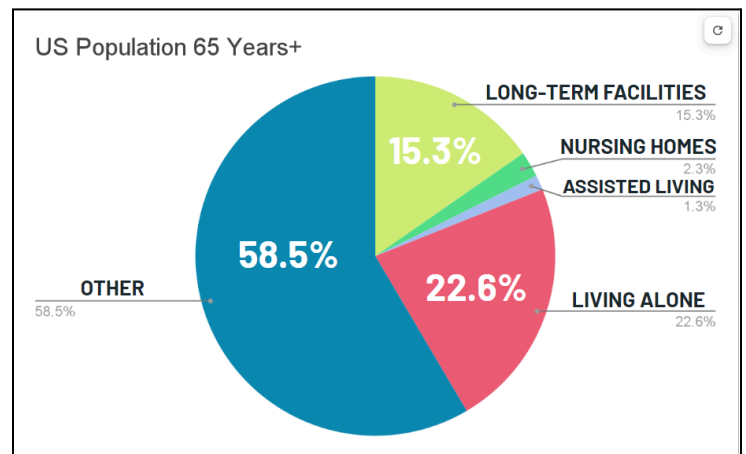
Positioning Statement

FamilyConnect is an eldercare manufacturer that strives to improve older adults' mental and social health. Unlike our competitors, we utilize artificial intelligence to enhance human interaction.

Markets

Our target market includes older adults and their caretakers, while our main end users are older adults in the US (primarily the one living alone or in long-term healthcare facilities). The pie chart on the right shows that roughly 40% of adults aged over 65 live apart from their family or completely alone; the FamilyFrame

would be incredibly helpful for them to connect with their distant friends and family. Additionally, since both the healthcare technology industry and older adult population are projected to grow exponentially in the future, so will our target market. The FamilyFrame will help older adults combat the digital divide and prosper socially and mentally.



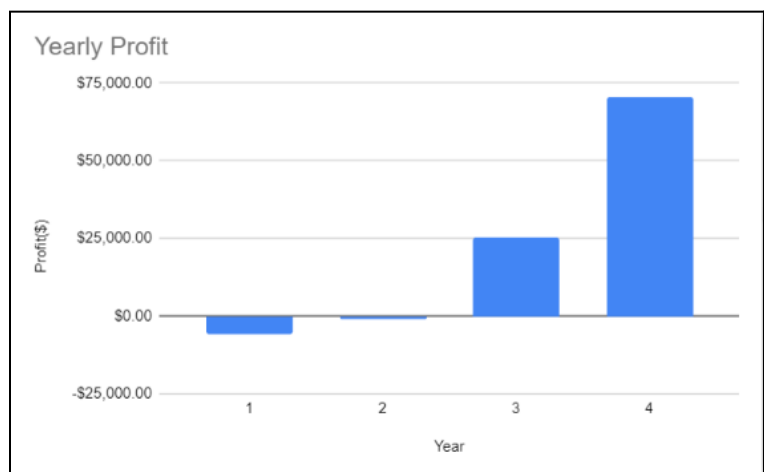
Value Proposition

The FamilyFrame offers a unique solution as a one-time purchase for the physical device and a monthly subscription for the AI cost. It serves as a communication and engagement tool to help older adults stay connected with their loved ones and comes with a 1-year warranty. Our product addresses the growing issue of social isolation and loneliness among older adults by providing a very user-friendly, AI-enhanced communication device that serves as both a digital picture frame and a conversation starter.

Financials

Our revenue sources stream from two main components: the physical frame/tablet and the monthly subscription. Our FamilyFrame sells for \$150 and costs \$58 to manufacture, yielding a ~61% gross profit margin. Additionally, our subscription sells for \$3 a month but only costs \$0.25 at max per month

to use the ChatGPT-3.5 Turbo API, yielding a ~92% gross profit margin. Our startup is roughly 2 years long, during which we will prioritize profiting from our app until we can comfortably mass-produce the physical FamilyFrame. We are projected to break even between Year 2 and Year 3, which is relatively short due to the low material cost of our app, easy accessibility to users, low competition, and high profit margins.



Conclusion

Our company, FamilyConnect, strives to do as our name suggests and connect families together across the growing digital divide. We want people to be as happy and healthy as possible and for nobody to be left out, which is why we target the elderly population with the FamilyFrame as their mental and social health are at the highest risk of all age groups.

We'd like to give a big thank you to all our mentors and helpers throughout the year to develop and improve the FamilyFrame. We'd like to thank Mr. DeBonis especially for connecting us to his developers and meeting with us frequently, as well as Mr. Smith, our Capstone Advisor and teacher, for his guidance and help throughout the entire project. It's been a long and difficult journey to get where we are today, but we find it incredibly rewarding thanks to Mr. Smith, Mr. DeBonis, and all our other project mentors.

As we head off to each of our own respective colleges this fall, we'll both reach out to professors and resources on campus and try to develop the FamilyFrame further. Hopefully, one day, FamilyConnect will really be connecting families all across the world for a happy, healthier society.

Citations

Articles

Fakoya, Olujoke A. "Loneliness and social isolation interventions for older adults: a scoping review of reviews." BMC Public Health, 14 Feb. 2020, bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-020-8251-6#:~:text=Approximately%2050%25%20of%20individuals%20aged,later%20in%20life%20%5B3%5D. Accessed 30 May 2024.

Malani, Preeti. "Trends in Loneliness Among Older Adults from 2018-2023." Healthy Aging Poll, University of Michigan, 13 Mar. 2023, [www.healthyagingpoll.org/reports-more/report/trends-loneliness-among-older-adults-2018-2023#:~:text=Measures%20of%20loneliness%20and%20social%20contact%20over%20time&text=More%20than%20one%20in%20three,8%25%20often\)%20in%202018](https://www.healthyagingpoll.org/reports-more/report/trends-loneliness-among-older-adults-2018-2023#:~:text=Measures%20of%20loneliness%20and%20social%20contact%20over%20time&text=More%20than%20one%20in%20three,8%25%20often)%20in%202018). Accessed 30 May 2024.

"Health Effects of Social Isolation and Loneliness." CDC, 26 Mar. 2024, www.cdc.gov/social-connectedness/risk-factors/?CDC_AAref_Val=https://www.cdc.gov/emotional-wellbeing/social-connectedness/loneliness.htm. Accessed 30 May 2024.

Images

Figure 1: "Suicide Statistics." American Foundation for Suicide Prevention, 11 May 2024, afsp.org/suicide-statistics/. Accessed 30 May 2024.

Figure 2: "The Elderly Population Is Growing Rapidly." Peter G. Peterson Foundation, 10 Apr. 2019, www.pgpf.org/chart-archive/0045_elderly-population-trends. Accessed 30 May 2024.